

RAMCO Institute of Technology, Rajapalayam

Department of Mechanical Engineering

Academic Year 2020 – 2021 (ODD Semester)

Innovative Practice(s) Description

Name of the Faculty : Mr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Subject Code and Title : OML 751 Testing of Materials
Year and Branch : IV year Civil, EEE and ECE
Topics Covered : Unit -II: Mechanical Testing – Brinell
Hardness Test
Name of the Laboratory : Strength-of-Materials lab
Website Link : <http://sm-nitk.vlabs.ac.in/exp10/index.html>
No.of Students participated : 60
Assignment given if any : Yes, virtual lab video making assignment
(If Yes share the uploaded
folder link)
Date : 12.08.2020

Demonstration of Brinell Hardness Test

OBJECTIVE

To determine the indentation hardness of mildsteel, cast iron, brass and aluminum

Brinell hardness test is most commonly used to test materials that have a structure that is too rough or too coarse to be tested using other test methods, e.g., castings and forgings. In Brinell testing machine the load is applied by a lever mounted on knife edges and carrying a hanger for suspending the required load.

The students can able to identify the material and learned about the experimental test procedure about to identify indentation of the material.

Course Outcome / Programme Outcomes	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12
CO2 Analyze engineering components using various mechanical testing procedure.	3	2	1	1	3	2	2	2	2

Brinell Hardness Test

Theory

Brinell hardness test is most commonly used to test materials that have a structure that is too rough or too coarse to be tested using other test methods, e.g., castings and forgings. In brinell testing machine the load is applied by a lever mounted on knife edges and carrying a hanger for suspending the required load. The supporting table for spacing the specimen can be raised or lowered by a steel screw by operating a large hand wheel. When testing, the load is applied and removed by hydraulic power controlled by a hand lever. The indenter used is a hardened steel ball which will have a diameter of 10mm, 5mm or 2.5mm. The reading microscopic has a 25-fold magnification. The gap between successive graduations of the scale is 0.5mm and a micrometer is arranged sideways, the eye piece of the reading microscope may be turned up by 90 degree so that the impressions of the ball can be measured in two perpendicular directions.

Diagram labels: Microscope, Indenter, Arm, Bending screw, Hand wheel, Lever, Hanger with loads.

BRINELL HARDNESS TEST

STEP 1 Diameter of the indenter $D = 10\text{mm}$. Select the load P based on the type of material selected (Mild steel, Cast Iron, Brass, Aluminum).

Selected material : Brass Load : 1000Kg

Material	Load P (Kg), diameter D (mm) (Diameter of the indenter $D = 10\text{mm}$)
Cast Iron & Mild Steel	$P = 30, D^2 = 3000\text{Kg}$
Brass, copper & bronze	$P = 10, D^2 = 1000\text{Kg}$
Aluminum, Magnesium & Zinc	$P = 5, D^2 = 500\text{Kg}$

(Each weight is equivalent to 250Kg, supporting hanging rod and base plate is 250Kg)

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STEP 2 Click on hand wheel then move mouse pointer over the hand wheel to rotate it in clockwise direction till the specimen is in position, after adjusting the specimen in specified position then move to the next step by clicking on NEXT button.

BRINELL HARDNESS TEST

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sm-nitk.vlabs.ac.in/exp10/index.html#

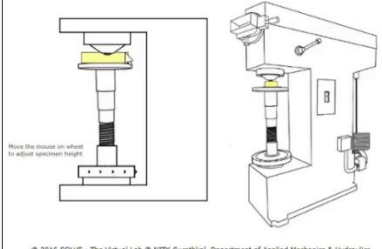
THEORYPROCEDURESELF EVALUATIONSIMULATORASSIGNMENTQUIZVIDEOREFERENCE

Simulator

BRINELL HARDNESS TEST

STEP 2 The specimen is placed on the supporting table, then the hand wheel below the table is turned in clockwise direction until the gap between the surface of the specimen and the indenter is 5mm.

TRIAL : 1



Move the indenter on wheel to adjust specimen height

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